



HW3

(Logic Synthesis for ROP3 & Enigma)

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Logic Synthesis for ROP3

- Try to synthesize HW1
- Compare performance between implementations
 - rop_smart
 - rop_256
 - Set **N = 32**
- Discussion
 - Which one is better
 - Why or why not
 - performance_index.txt



Logic Synthesis for enigma

- Try to synthesize HW2 part4
 - Remove _part4 from filenames & module names
- Performance of your circuit will be measured with PI
 - $PI = \text{Area} * \text{timing}$
 - e.g. area = $20000\mu m^2$, timing = 5ns
 - $PI = 1 \times 10^5$
- PI ranking list
 - Your grade of HW3 depends on your rank among students
- Note:
 - Pass all simulations
 - one output per cycle
 - Need to be synthesizable and no latch in the netlist except for gated clock
 - Testbench will be given after deadline of HW2



Grading Policy

- Logic Synthesis for ROP3
 - 10 points
- Logic Synthesis for Enigma
 - 10 points if you fill in PI correctly in the PI ranking list
 - 80 points PI ranking if your $PI < 1 \times 10^5$
 - $Score = 80 \times \left(\mu + \sigma \times \Phi^{-1} \left(\frac{r-0.5}{N} \right) \right) \%$
 - $\mu = 60$
 - $\sigma = 20$
 - Φ^{-1} : inverse of CDF of standard normal distribution
 - r: your rank
 - N: # of students in ICLAB
 - Additional 2% & 1% of semester grade will be given to top1&top2 students



PI ranking list

- Submit your PI with google forms
 - Log in with your **GAPP account registered in the beginning of the semester**
 - Please fill in your PI correctly
 - anonymous, choose your name to be shown in PI ranking list freely
 - You will receive 10 points if you correctly fill in your PI value in the ranking list.



Deliverable

HW3_{studentid}\

```
└rop3
├ performance_comparison.txt
├
├ └hdl
├   rop3_lut256.v
├   rop3_smart.v
├   spyglass_lut256.rpt
├   spyglass_smart.rpt
├
├ └report_lut256
├   report_area_lut256.out
├   report_power_lut256.out
├   report_time_lut256.out
├   syn_rop3_lut256.log
├
├ └report_smart
├   report_area_smart.out
├   report_power_smart.out
├   report_time_smart.out
├   syn_rop3_smart.log
├
└─enigma
├ performance_index.txt
├
├ └hdl
├   enigma.v
├   spyglass_enigma.rpt
├   {xxxx}.v
├
├ └sim
├   rtl_sim.f
├   run_rtl_sim.sh
├   test_enigma.v
├
├ └syn
├   0_readfile.tcl
├   report_area_enigma.out
├   report_time_enigma.out
├   synthesis.tcl
├
└
```



Enigma

- Deadline: 10/23 11:59p.m.
- Submit to eeclass
 - Make sure the file delivery and organization meet the requirement
 - 10 points deducted from this homework for wrong file delivery, wrong file organization, or wrong file naming
 - Delayed submission: $grade_{delayed} = grade \times 0.9^{delayed_{days}}$
- Q&A
 - Feel free to send e-mail to TA (紹華) or post your question on eeclass
 - Do not show your code or email your code to TA. Coding by yourself !
- Using generative AI is not recommended.
 - If used, please explain in the report how it was applied